AME 485 – Aerospace Structures

Project 6

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Part 1 #1

10 Lessons I Have learned – Different approach

1. Mistakes happen often, make sure you learn from the mistake and own up to the mistake if you personally made it
2. Life lesson that Vinay taught me – Never say more than is needed, for example, on a resumé or in an interview only say or do what is required
3. Always remember the lesson
4. Be compatible with your groupmates, peers, and coworkers – Be open to suggestions, ideas, and other individuals, they can help you grow as an engineer
5. Learn how to work with a team and learn each others' strengths and weaknesses to build effective communication. This will also enhance productivity and efficiency
6. As an engineer always carry pocket margin. Even though we have the factor of safety an extra "pocket" margin is always good to have in case a mistake occurs
7. Always have peers review your work or other peers/management check your work.
8. Have a work life balance and understand your own strengths and weaknesses
9. Enjoy the accomplishments when they happen and spend time understanding the failures
10. The Aerospace industry is massive and requires a lot of continuous parts and teams, be able to communicate effetely and efficiently

- Flexing communication is essential because you need to adapt to your work environment. Being able to adjust your communication style to one that is not your own is a critical skill, especially in industry. For example, if you have a boss who dislikes colorful designs, prefers straight-to-the-point communication, and wants everything presented as quickly as possible, your presentation at the next work meeting should align with those preferences. Even if you are a colorful, artsy, and detail-oriented person, if your boss values concise bottom-line results without explanations, you need to adapt and meet their expectations. This is one example of tailoring your presentation or verbal communication skills to a specific individual (in this case, your boss). Similarly, you may need to adjust when working with a data-oriented team or person, a creative team, an introverted coworker (where you may need to be more charismatic), or a big-picture thinker. Adapting to these different scenarios is crucial to maximizing productivity, efficiency, and managing emotions.

Part 1 #2

Review JSC 65828B

1. Determinantal Deformation – A requirement that flight hardware structure should have no detrimental deformation when exposed to limit loads multiplied by yield factors of safety during its service life. (I.e. hardware should not deform when subjected to the highest expected loads + F.S.)
    1. The Rational is that there shouldn't be structural deformation that compromises functionality or safety. It can lead to the following:
        - Misalignment, contact, or interference b/w components
        - Internal load redistribution, affecting structural integrity
        - Exceeding space or design envelops
        - Reduce strength, lifespan, or performance of the structure
        - Degradation of thermal protection or coatings
        - Equipment malfunction or personnel risk
        - Loss of aerodynamic or functional performance
        - Reduced confidence in flightworthiness
        - Induce leakage above specified rates
2. Creep Life – Flight hardware structure should have adequate structural life with a creep life 4x (times) the service life
    1. The rationale is flight hardware should be designed to prevent damage from creep during its service life. The material selection and approach used to assess creep should be documented in the Structural Verification Plan (SVP). Creep life can be evaluated using appropriate methods, such as the time-fraction rule, to determine total damage.
3. Doors and Hatches in Habitable Modules – These modules should withstand applicable loads with the doors or hatches both open and closed for the applicable ground and mission environments.
    1. The rational is that the integrity of the habitable modulee must be maintained throughout all phases of service life for all hardware configurations, including when the hatches are open or closed.

Review JSC 65828B – Cont.

4. Critical seal test ports – Assemblies containing critical seals with a major outer diameter greater than 6 inched shall include leak test ports and conductance grooves within the seal interstitial area to accommodate redundant seal verification (I.e. critical seals must include tests ports and grooves to enable redundant testing for leaks)
    1. The rationale is each seal individually needs to be tested so large grooves and tests ports are needed.
5. Design loads for collapse – Loads are treated as ultimate loads (with F.S.), except for loads that stabilize the structure against buckling. If the load alleviates buckling, then the ultimate factor of safety is not applied to that component
    1. The rationale is that applying a factor of safety to a load component that tends to alleviate buckling will result in an unconservative assessment of buckling load capability (may cause an overedging/heavier design)
6. High-Cycle Fatigue Life – Flight hardware that is subjected to high-cycle fatigue will have a minimum fatigue life of 10x (times) the service life
    1. The rationale is that high cycle fatigue can propagate flaws to failure. For example, the vibration loading of a turbine blade due to structural resonance – overtime this can grown flaws (ex. microcracks) into larger failures.
7. Fasteners and Fastened Joints – Structural joints and fitting must include a fitting factor of 1.15 in the design analysis, applied to both limit and ultimate loads conditions for all phases of service life.
    1. The rationale is that the fitting factor is used to account for potential variations in internal load paths within the actual structure that are not covered by idealized analytical models.
8. Mechanical and Thermal Factors of Safety – The factors of safety and test factors must be applied to both mechanical and thermal stresses and loads.
    1. The rational is that most safety factors apply to mechanical loads (I.e. inertial, aerodynamic, random vibration, shock, crew-induced). Thermal loads, derived from less certain, unverified, and often late-stage models, use the same safety factor due to their uncertainty. A lower factor of safety could be allowed if, the thermal data used for structural design can be demonstrated to have a low level of uncertainty.

Review JSC 65828B – Cont.

9. Design Loads for Rotating Machinery – The design will consider all applicable loads over the service life including self-induced loads due to operation.
    1. The rationale is that the design load case definition should consider all relevant loads and load cycles over the service life as well as self-induced conditions due to operation.

10. Pressurized Hardware Design Requirements – Pressurized hardware must remain dimensionally stable to ensure functionality of structural attachments, pressure connections, and openings for doors or hatches throughout their service life in the applicable environments.
    1. The rationale is to ensure successful function and operation of structural and mechanical attachments, pressure connections, and openings for doors and hatches by taking into account worst-case scenario dimensional variations over the service life environment. This requirement also applies to habitable modules.

Part 1 #3

3. The purpose of structural qualifications is to make sure the structure is defined via the correct design qualifications and hardware acceptances. The qualifications require the following needs:
   - Designs qualified to account for part-to-part variability due to assembly procedures, build-to-build tolerances, and uncertainties in redundant load paths
   - Flight hardware accepted to screen out poor workmanship or preclude a hazardous conditions
   - This verification process needs to encompass and consider all potential failure modes and all potential load conditions (I.e. detrimental deformation and/or dynamic launch loads) (There are several other potential failure modes and load conditions.)
   - These verifications are defined based on the loads found in testing, CFD, or other tests to determine loads and failure modes
   - The qualification process provides verification of the workmanship process, provides qualification margins above limit loads, and provides an approach to anchor analysis methods.
     - An example of this qualification process is to test article representations of all available production units that have undergone flight-like processing to then determine if the article or other structures using the same article can be furthered qualified.

- The purpose of acceptance proof testing includes the following:
  - It verifies workmanship
  - Required for composites and bonded structures
  - Nondestructive evaluation is recommended after proof test
  - An example is, all flight tanks and solid rocket motors are proof tested
  - Qualification testing and design analysis only verify design adequacy so proof testing is needed
  - As built hardware is usually imperfect (variability)
    - Manufacturing defects
    - Damage during handling and shipping processes

#3 Cont.

CONT of acceptance proof testing, last slide

- If hardware can not meet spec
  - Either accepted by analysis or testing or redone
  - Validations become important, either reject or repair
  - Verify the as-built hardware will function as designed

- The purpose of analysis is to predict limit stresses, determine a factor of safety, and understand the materials mechanical properties. Analysis allows for the analytical margin calculations to determine the structural capability under nominal and dispersed operating conditions.
  - This is important because it helps identify critical margin conditions for various failure modes
  - Understand whether a critical failure mode is fully tested
  - Understand whether the relied upon hardware variability is accurate enough for the conditions or project

Part 1 #4

4. The factor of safety covers for beyond the expected loads or stresses of what a structure or material will encounter in normal operating conditions. The factor of safety accounts for the following:
   - Variations in workmanship or hardware manufacturing variabilities
   - Uncertainties with materials and their mechanical properties
   - Uncertainties in load and stress estimations in operating conditions
   - Environmental factors, wear, and fatigue
   - NOTE: The factor of safety does not justify a “no-test” approach. If there is a “no-test” approach the factors of safety tend to be higher to account for mistakes/variability.

Part 2 #1

- The design process for an aircraft begins with criteria, objectives, and different requirements, which for a passenger aircraft include:
  - Passenger count
  - Service life
  - All weather conditions
  - Damage tolerance
  - Reliability and repairability
  - Range, speed, and avionics

- After the criteria and requirements are found, the configuration is set and the external loads and environments are accounted for, which include for a passenger aircraft the following:
  - Pressures, loads, and stresses
  - Inertia, thermal, and acoustic loads
  - Adverse environments

- The internal loads and load paths are found, then analysis occurs which includes various testing methods where the allowables for the aircraft are found. Then the aircraft is certified.
- An overview of the design process includes
  - Problem description
  - Preliminary design
  - Analysis
  - Manufacturing
  - Final aircraft delivery

Design improvements occur here (iterations)

Part 2 #1

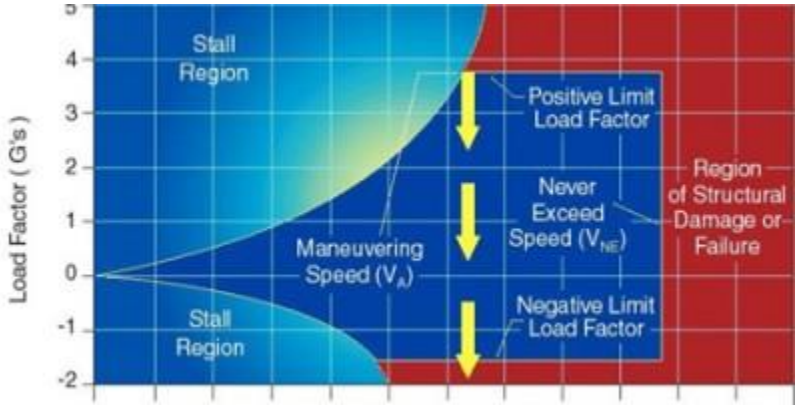
CONT. Part 2 #1

- The source of loads and types of loads that must be considered include the following when designing an aircraft: These are some major examples of the sources for loads,
  - Airplane wing body junction
  - Bird strike or blade impact
  - Crack growth and residual strength
  - Airframe stress
  - Static, dynamic, and weight analysis
  - Structures integration
  - Environment (wind gust, rain, hail)
  - Buffet, inertia, vibration, maneuver, and deflection flight loads
  - There are also ground loads and other loads such as braking, towing, jacking, bumps, etc.

- Types of loads include:
  - Fuel slosh
  - Ultimate and fatigue loads
  - Floor and frame loads
  - Pressure loads
  - Miscellaneous loads (tire bursts, towing, on ground maneuvering)
  - Load carrying doors
  - Landing gear
- A good example of source and type loads used in class is, “a crack causes a hole somewhere on the aircraft, then a sudden depressurization occurs”. The aircraft needs to be analyzed for a depressurization to make sure it can manage the loads.

Part 2 #2

2. A V-N diagram is an airspeed and load factor plot. On the X-axis, the load factor is represented and on the Y-axis the airspeed is represented. The diagram is divided up and indicates crucial points when designing aircraft such as the positive and negative limit loads, the ultimate load factors, the proportional limits, and the stall region. It is important because, a V-N diagram allows for engineers to determine the minimums and maximums in aircraft design. You have to make sure the design is made to cover for the "Corners of the box". Engineers don't want a stall or structural failure to occur, and the diagram provides a big picture on what happens when airspeed changes and how that effects load factor.

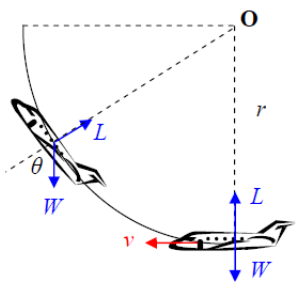


[1] Name, "The V-G diagram (V-N diagram)," Aviation Performance Solutions, https://apstraining.com/resource/the-v-g-diagram-or-v-n-diagram/ (accessed Dec. 4, 2024).

Part 2 #3

3. Done by hand on iPad, the derivation is shown below.

Level Turn: Pull-Up



Load factor

$$\begin{cases} L = nW = \frac{1}{2}\rho V^2 S C_{L,max} \\ n = \frac{L}{W} \end{cases}$$

$$L - W = \frac{mV^2}{r}$$

$$L = W + \frac{mV^2}{r}$$

$$n = \frac{L}{W} = \frac{W + \frac{mV^2}{r}}{W} \qquad W = mg$$

$$n = \frac{L}{W} = 1 + \frac{\frac{mV^2}{r}}{mg}$$

$$n = \frac{L}{W} = 1 + \frac{V^2 r}{g}$$

$$\boxed{n = \frac{L}{W} = 1 + \frac{V^2}{rg}}$$

Part 2 #4

3. Done by hand on iPad.

$W = 100{,}000$ lbs

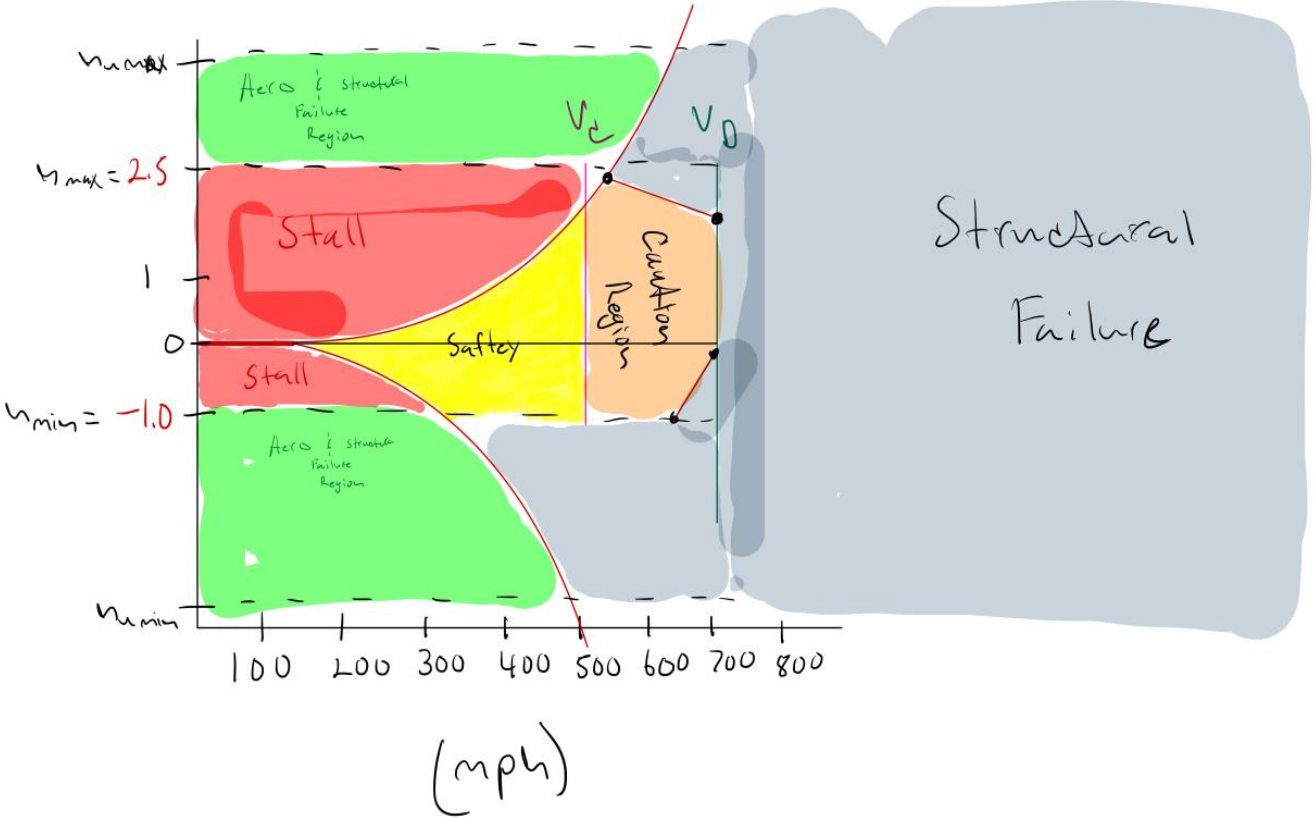
$(S) = 270\ ft^2$

$C_{L\,min} = -1.0$

$C_{L\,max} = 1.6$

$V_c = 500$ mph $\rightarrow 255$ m/s

$V_d = 1.4\,V_c = (1.4)(500) = 700$ mph $\rightarrow 312$ m/s

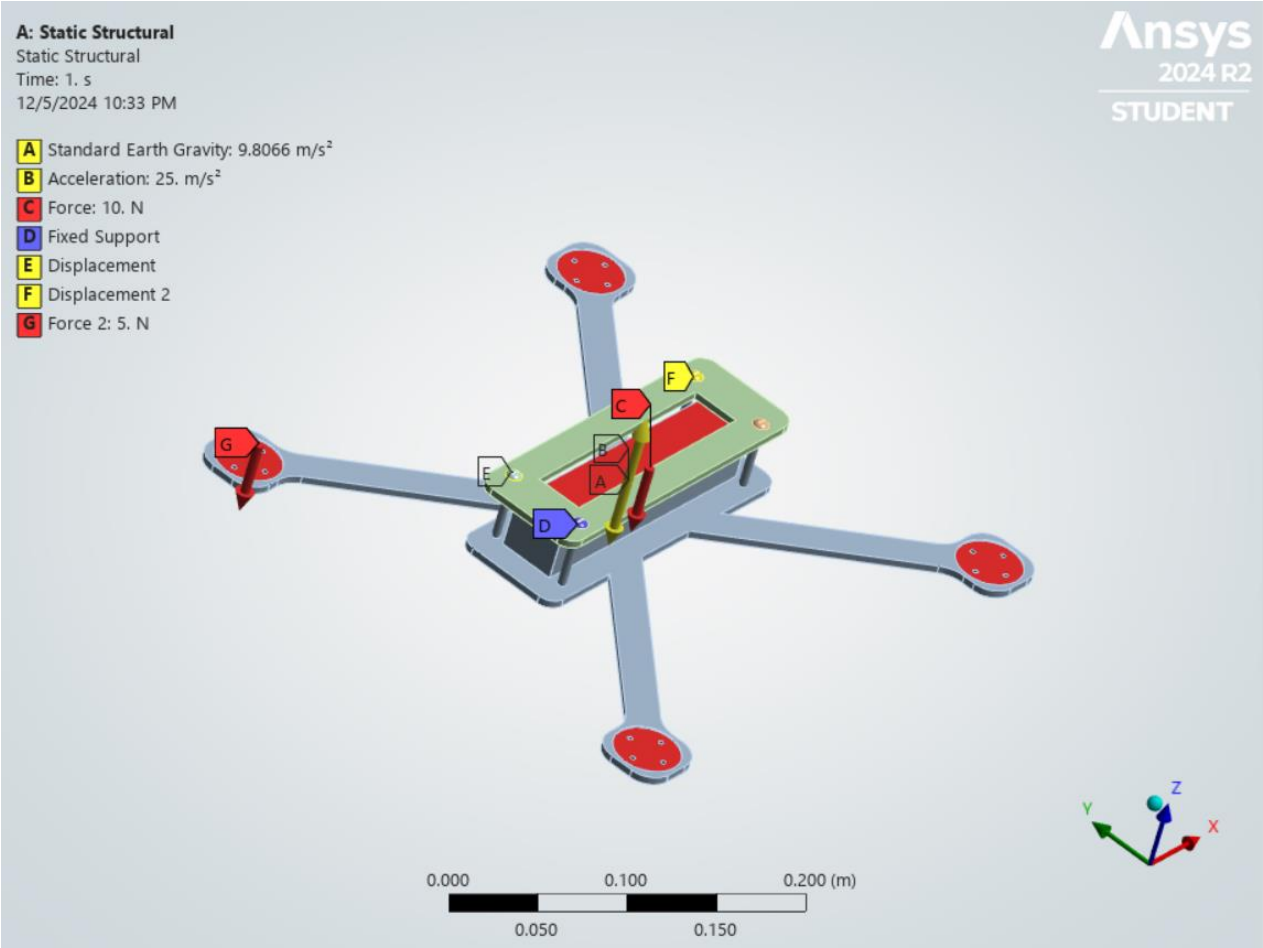


Part 3 ANYS

Did not work but tried my best

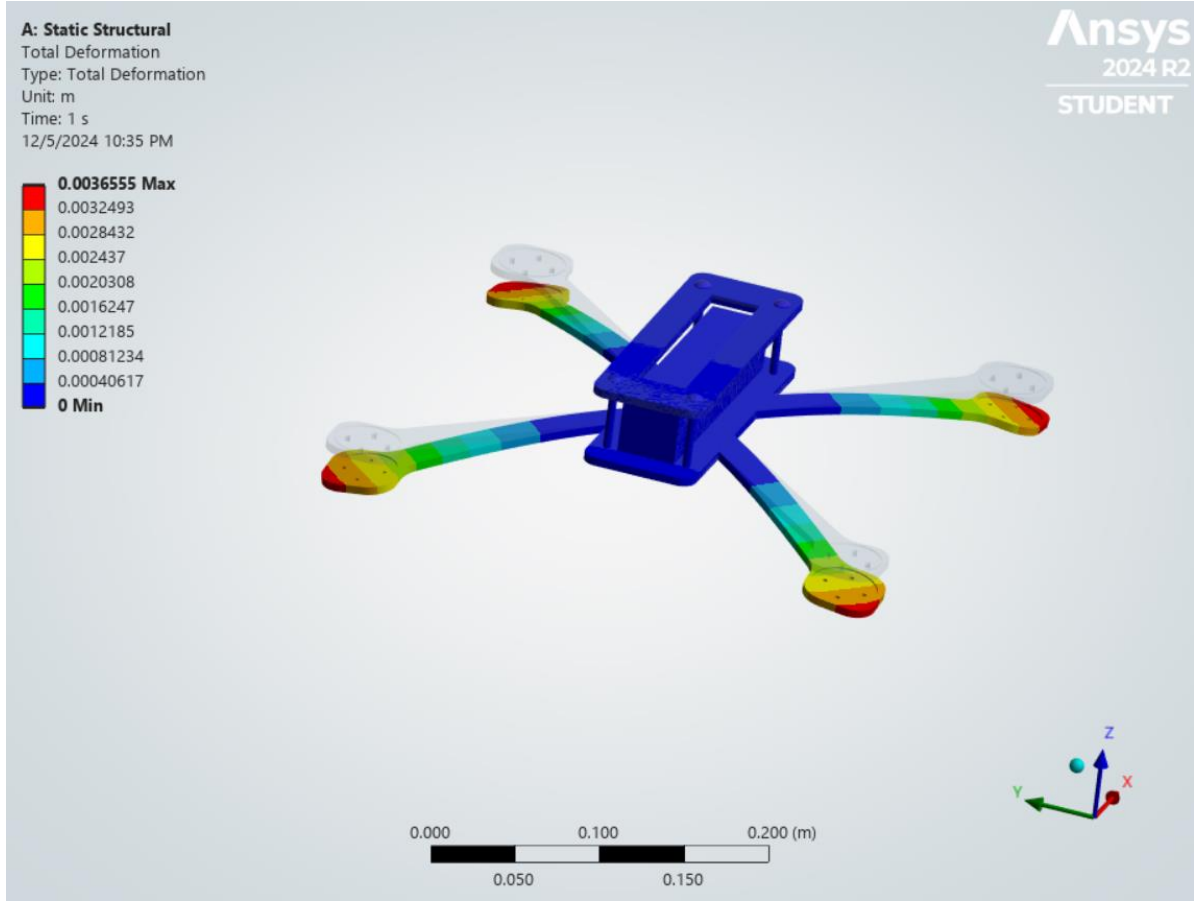
The acceleration, fixed support, force, displacement, and earths gravity were all added properly to the model.

Following the video tutorial, the model was first tested using ABS plastic, after switching to Al 6061-T6, the model failed, and no further analysis could be done.



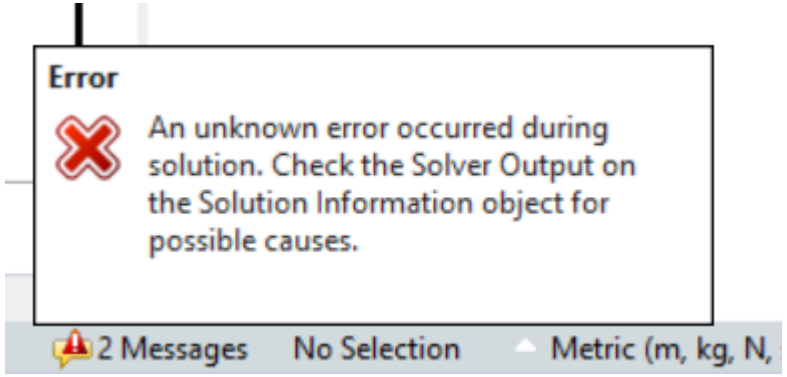
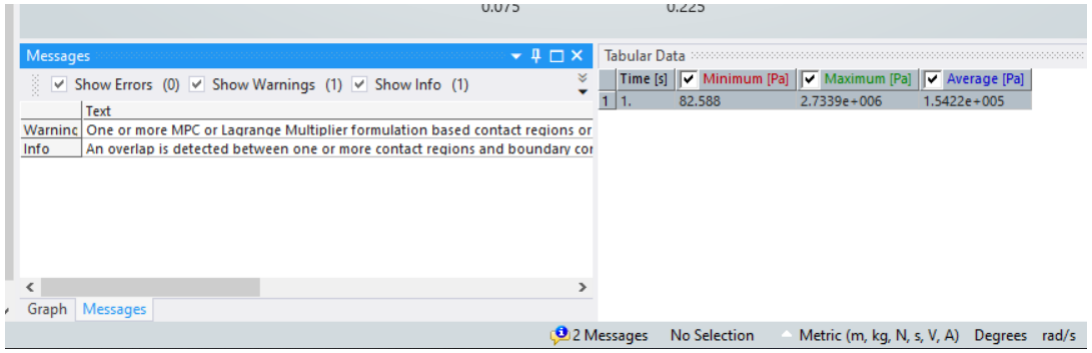
Part 3 ANYS

Did not work but tried my best



For ABS plastic, the total deformation and equivalent stress are shown above in the following pictures.

Part 3 ANYS

Did not work but tried my best

With Aluminum alloy 6061-T6, the ANSYS model did not work and gave me this error. Thus, I was not able to discuss the factor of safety and A-basis allowable question.